

Case Study

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Background

Introduction

This lesson provides an opportunity to analyze a business problem and create a comprehensive solution to that problem.

Lesson objectives

This lesson is designed to help you to:

- Analyze a business problem
 - Create a comprehensive solution to solve the business problem
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Case Study

Overview This Case Study is based on a real implementation and is designed to test your learning in class. The Case Study is planned to take 8 hours (It is intended for flexibility on Friday or earlier if applicable). Because of the complexity of this study, regular stopping points are inserted to bring back the class together for discussion and progress checks. Those falling behind will be brought current by the instructor providing the solution to missing or non-working components.

Timeline The following timeline is a guideline established to assist you in managing your time for completing the Case Study. Adjust timing accordingly to pace of class. Examples being starting Fourth Day or Starting 5th Day but going a little longer.

Fourth Day:

1:00pm - 2:00pm - Discuss and explain the case study; Q&A time lines for the exercise and what services/adapters are required.

2:00pm - 4:00pm - Steps 1-5 from “Steps for Completing the Case Study”.

4:00pm - 4:30pm - Breakpoint #1 - Class discussion and debugging of file directories and database table.

Fifth Day:

9:00am - 11:30am - Steps 6-9 from “Steps for Completing the Case Study”.

11:30am - 12:00pm - Breakpoint #2 - Class discussion and debugging of ErrRtn, and Output processes, and services/adapters.

12:00pm - 1:00pm - Lunch

1:00pm - 2:30pm Steps 10-13 from “Steps for Completing the Case Study”.

2:30pm - 3:00pm - Breakpoint #3 – Class discussion and debugging of Main.

3:00pm - 3:30pm - Final Review/Course Wrap-up/Evals.

The Business Problem

Overview

Historically, the company is operated as two autonomous divisions; one for trucking and logistics, and one for pharmaceuticals. Recently, Sterling B2B Integrator has been purchased and installed, and planned to unite processing into a single system through it. The backend systems will remain independent for the time being; still the evaluation of which ERP software to choose and implement in the future is under discussion.

Company receives inbound data in EDI (X12) format, from Sterling Collaboration Network (our VAN of choice). Some of the data is intended for the company pharmaceutical division, some for the logistics division. Pharma orders must be written to the ORDERS table in the database. Logistics documents must be sent to the logistics application, which requires the data be in EDI format, but with a specific set of delimiters that the app system needs and which the partner will not provide. For data warehousing purposes, must write a copy of the data to a directory, in XML format.

Pull the data from the VAN by way of a communications package. It is being done through FTP communications and it is already established. The files will be transmitted to a folder on the B2Bi server.

Error handling is required, as is hands-free processing.

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The Business Problem

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Bootstrapped FSA Config Vs. FSA in a BP and CLA2

During this case study the business flow and hints will have you build a business process with a FSA and an Invoke service to start off your process flow, and add error handling if there are no files to collect. Then in the last process you will add a CLA2 to run a script to clear the collection directory.

If limited on time, build a bootstrapped, scheduled FSA configuration as your “starting BP. Then remove the requirement for the CLA2 because the FSA will delete the file on collection

If you have more time try the FSA BP and CLA to clear the directory. Why would someone build all the extra to collect and delete a file? By having a FSA in a BP it gives you more control of the configuration. Such as setting false to “Delete on Collect” and having another service delete the file once processing was complete.

Setting an error on “No files to collect” is not very common as usually it is desired for the FSA to constantly check for incoming files and multiple errors would fill up error reports/logs. This error would be beneficial though, if files were expected at a specific time and if they did not arrive someone should be notified.

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The Business Problem

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The Business Flow

The Start business process collects data from a directory with an FSA (scheduled). When data is collected, bootstrap Main. If no data is present for collection, invoke ErrRtn business process. This could also be done with a bootstrapped and scheduled FSA configuration instead of a BP but you would lose the error handling.

Main accepts the data handed to it by Start, and encodes it into XML, placing it into ProcessData. Test the data; if it is for the Pharma division, write the PO number, and customer number to the ORDERS table. If it is for the logistics division, change the segment terminator from a "*" to a "=" (equal) and write the altered EDI data out to a processing directory. After this processing is done, pass the results to Output business process. If there are errors anywhere along the line, invoke ErrRtn.

Output accepts the data handed to it by Main, and invokes two paths at the same time; one will call a script outside of Sterling B2B Integrator that will clear the inbound directory, The other moves the XML in Process Data to Primary Document, and write it to a data warehousing directory. If there are errors anywhere along the line, invoke ErrRtn.

ErrRtn is called from multiple processes, as a generic error handling process. Project specifications require that errors be reported to the Sterling B2B Integrator admin via email, requiring the use of an SMTP Send Adapter.

Pre-requisites to Complete the Case Study

Introduction

In order to complete this case study, you will require the following:

1. File System Adapter (2)
2. OnFault activity
3. Invoke Business Process Service
4. XML Encoder Service
5. Choice Start/Choice End
6. Lightweight JDBC Adapter

In order to extract the correct values for use in updating the database in the lightweight JDBC Adapter, you will need to know where the PO_Number and Customer Number are located. If you look at your map you will see the following:

- PO Number = Element 0324 under the BEG segment
- Customer Number = Element 0093 under the N1 segment in the first loop of the Group N1.

Here is an example of an XPath statement finding the Customer number
`//N1[1]/_0093/text()`

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Pre-requisites to Complete the Case Study (Continued)

Introduction (Continued)

7. Document Keyword Replace Service

Important!	The Document Keyword Replace Service is not part of your course materials. However, all service and adapter documentation is available on Knowledge Center. There is no default configuration. One must be created before it will show in the GPM
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8. Assign activity

9. Command Line Adapter

Command Line Program: Batch program is supplied in lab files folders.
Program name is Clear_Dir.sh, and will delete all files in directory
c:\Inbound_from_Comms.

10. All Start/All End

11. SMTP Send Adapter

12. Create a table in the Sterling B2B Integrator database to the required specifications

13. Access to product documentation

14. XPath to conduct a choice

15. XPath for DOMToDoc

Case Study Hints

Introduction	The following pages contain some tips that can help you as you work through the case study.
Additional Information	1. The Start business process contains a File System Adapter and an Invoke Business Process stencil. The FSA should read from Inbound_from_Comms and pass the document to the Main business process. 2. Create a business process called ErrRtn that uses an XML Encoder Service and an SMTP Adapter to send an e-mail (optionally also a note added to Process Data) that there is an error in the business process instance. Use the XML Encoder and xpath to pull from Process Data the Error tag contents and insert into the e-mail. Can also be done with a DOMToDoc. 3. Capture all errors with an OnFault that calls ErrRtn via an Invoke Business Process. 4. Create a map for translating EDI to XML. This map must work for both the Ellis 850 documents and the Unlimited 216 documents. A suggestion is to create an 850 input side map first, then add the 216 segments not there already (PUN, G61, TEM) to the 850 map. Since the map is being used for conversion to XML, data not in the document will be ignored at this step. Here the XML Encoder map will just convert the data it receives. Use 3040 version as previous maps. 5. Create in Main an XMLEncoder stencil, choose to Encode non-XML documents, and use the map you created. a. Set edi_input_element_delimiter to 2A * . b. Set edi_input_segment_delimiter to 7E ~ . c. Set map_name to your combo map. The delimiter settings will allow the map to separate the segments and elements into individual tags instead of grouped together into one tag by the segment. This will make writing your rules and assigns much easier.

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Case Study Hints

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Additional Information (Continued)

6. Once the document has been converted into XML in Process Data, create a rule that tests whether the data is from Ellis Labs (N1 contains “ST” and “Ellis”) or from Unlimited (G61 contains “SH” and “CHRISTIAN”). If it’s for Ellis (Pharma), use Assign activities to pull the PO_Num and CustNum values out of the document and into their own Process Data tags. Using a Lightweight JDBC Adapter, write an insert statement that pulls the PO_Num and Cust_Num values from Process Data and inserts them into the table created earlier as CUST_ID and PO_NUM.
7. If the data is for Unlimited (Logistics), call the Document Keyword Replacement Service and replace ~ with = as the segment delimiter, and use an FSA to extract the document to the Altered_Delimiter directory.
8. After the Choice End, call an Invoke Business Process to invoke the Output BP.
9. Output uses an All Start to enact two different paths at the same time. One path uses a Command Line Adapter to call the Clear_Dir.bat script to clear Inbound_from_Comms. The other path uses an XML Encoder to convert Process Data to Primary Document. Use a File System Adapter to write the Primary Document to the Output directory. Configure an OnFault to handle any errors.
10. It is very common in the real world to run business processes to see how things will align in Process Data in order to write correct XPath statements. You do not have to but an example would be to run your map to see the tags and data from the EDI document to write your rules and assigns.

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Case Study Hints

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11. If you do not set delimiters for your map the following assigns can be helpful in pulling out pieces of information:

a. On an Assign: `substring(//_0353/text(), 8, 4)`

It reaches into ProcessData, locates element _0353, counts from the left 8 bytes, and extracts the next 4. Result: The PO number from the data in ProcessData for use in the adapter is extracted.

b. On an Assign: `string-length(//_0098/text())`

It reads the overall length of the element in ProcessData and stores that number of bytes as an integer for future use.

c. On an Assign: `substring(//_0098/text(), 5, CustNumLen)`

Here, CustNumLen might be the To value from the last Assign. It reaches into the element in ProcessData, places a pointer at the fifth byte, and extracts the rest of the element for use in the adapter.

Completing the Case Study

Overview	In order to successfully complete the Case Study, you must complete the following steps in their given order. 1. Analyze the business problem, plan/sketch a solution. 2. Gather requirements (directories to be created, email address, mail host/port, command-line script location, and function, database table schema). 3. Develop three file system directories (Inbound_from_Comms, Altered_Delimiter, Output_Data). 4. Create database table ORDERS within the Sterling B2B Integrator database. 5. Build the business processes.
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To Create the ORDERS Database Table
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Database Schema: Table is to be added to Sterling B2B Integrator database in class.
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Table Name: ORDERS

Fields: CUST_ID (string 20), PO_NUM (string 10)
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Creation Instructions:

- | |
|--|
| <ol style="list-style-type: none">a. Go to Operations > System > Support Tools > SQL Manager in the UIb. Type create table ORDERS(CUST_ID char(20), PO_NUM char(10)) and press Enterc. Check your work by typing describe ORDERS and click Execute |
|--|

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Completing the Case Study

(Continued)

**To Create the
ORDERS Database
Table
(Continued)**

6. Configure services and adapters Create and check-in ErrRtn.
 7. Test/debug ErrRtn.
 8. Create and check-in Output.
 9. Test/debug Output with and without errors to call ErrRtn.
 10. Create and check-in Main.
 11. Test/debug Main with and without errors to call ErrRtn.
 12. Manually test entire process.
 13. Set schedule for Collection side of FSA or schedule Start BP to test automation.
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Deliverables Provided by Instructor

Overview

Your Instructor will provide you with the following information to assist you in completing the Case Study.

1. Database table schema.
2. Command-line program.
3. Two EDI interchanges; one for Pharma and one for Logistics.
4. Mail server documentation
5. Documentation on this case study, with breakpoints for discussion.
6. A completed version of this, tested and demonstrable, for discussion and to catch up those lagging behind.

Additional Information

The case study that provided in this lesson is intended to challenge you with a real-world business problem. The examples hear are one way to solve the problem. Feel free to experiment if desired.

If you have any questions during the Case Study or you need assistance, contact the instructor.

Lesson Review

Completed Objectives

This lesson was designed to help you to:

- Analyze a business problem
 - Create a comprehensive solution to solve the business problem
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